THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SOCIAL SCIENCE CHAPTER TEST SET A

Chapter 4: **Atmosphere and Climate** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat maps and diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The atmosphere is held around the Earth by: 1

(a) magnetism **(b)** wind
(c) gravity **(d)** rotation of the Moon

2. The layer of the atmosphere where weather phenomena occur is the: 1

(a) mesosphere **(b)** stratosphere **(c)** thermosphere **(d)** troposphere

3. The thermosphere extends from about: 1

(a) 80 to 700 km **(b)** 0 to 12 km
(c) 12 to 50 km **(d)** 700 to 1000 km only

4. The average weather of a place over a long period is its: 1

(a) climate **(b)** weather **(c)** temperature only **(d)** rainfall only

5. Moisture in the air in the form of water vapour is called: 1

(a) insolation **(b)** humidity **(c)** pressure **(d)** precipitation

6. A local wind blowing from land to sea at night is a: 1

(a) land breeze **(b)** monsoon **(c)** trade wind **(d)** sea breeze

7. During summer the Indian subcontinent has: 1

(a) high pressure over land
(b) low pressure while the ocean has high pressure
(c) same pressure everywhere
(d) no pressure difference

8. The retreating monsoon season lasts from: 1

(a) June to September **(b)** March to May
(c) January to March **(d)** October to December

9. The Indian Meteorological Department (IMD) is responsible for: 1

(a) weather forecasting **(b)** elections
(c) printing currency **(d)** census

10. A rain gauge measures: 1

(a) humidity **(b)** pressure **(c)** rainfall **(d)** wind speed

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 10–15 words each.

11. What is the atmosphere? State two of its functions. 2

12. Name the five layers of the atmosphere from the Earth's surface. 2

13. Why is the troposphere important? 2

14. What is the importance of the ozone layer? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Describe the composition of the atmosphere. 3

16. Explain the importance of the atmosphere to life on Earth. 3

17. Describe the troposphere and the stratosphere. 3

OR

Describe the mesosphere and the thermosphere. 3

18. Explain how latitude and altitude affect temperature. 3

OR

How does distance from the sea affect the climate of a place? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) Describe the structure of the atmosphere with a diagram.
(ii) Give one feature of each layer. 5

OR

Case study: A passenger aeroplane flies at about 11–12 km above ground where there are no clouds. 5
(i) Which layer is this?
(ii) Why is it suitable for flying?
(iii) What is the lower layer called?
(iv) Name the layer which shields us from UV.

20. (i) Distinguish between weather and climate.
(ii) List the elements of weather.
(iii) Explain how temperature varies with latitude and altitude. 5

OR

Case study: Table: Station A rainfall is 183.4 cm per year with 61 cm in July; Station B has 0.2 cm in January and 4.5 cm in April. 5
(i) Which station is wetter?
(ii) In which season does A receive most rainfall?
(iii) What causes it?
(iv) Suggest the crop pattern for A.

END OF QUESTION PAPER — Total: 40 Marks